Overview of current text-to-speech techniques: Part II — prosody and speech generation

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Speech synthesis is an increasingly important technique in interactive voice systems, and is crucial to the next generation of spoken language dialogue systems. Described here are many of the techniques and algorithms used in modern commercial text-to-speech synthesisers, where real world issues of robustness, reliability and processing cost compete with the specific requirements of naturalness and intelligibility. This paper concentrates on using the results of the linguistic analysis of the input text to generate an appropriate synthetic prosody and speech signal. The importance of the rhythmic characteristics of synthetic speech is emphasised by placing the prosody generation task within the framework of metrical phonology.

1. Introduction

The conversion of text to speech (TTS) has been described [1] as essentially a two-stage process (see Fig 1).These are an analysis stage,which derives a linguistic structure from the input text,and a generation stage,which uses the linguistic structure to synthesise the speech,including the generation of intonation,rhythm,etc.

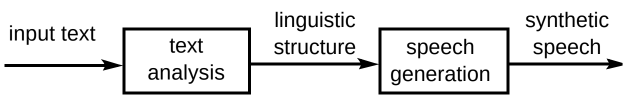

Fig 1 Text-to-speech process

The result of the text and linguistic analysis stage [1] is detailed linguistic information about the structure of the input text, in particular, syntactic, lexical, phonological and some semantic information. The following sections of this paper describe how the linguistic structure is used to generate the synthetic prosody (section 2), and the final speech signal (section 3).

2. Prosody generation

The biggest single problem confronting TTS systems remains their poor generation of prosody for unrestricted text. The term ‘prosody’ covers a wide range of features characterising the ‘musical qualities’ of speech, including phrasing, pitch, loudness, tempo and rhythm. These features are to some extent interrelated, but are often considered independently, placing conflicting demands on the ideal of a single prosodic structure. In practice, deriving all the relevant features within a unified framework is seldom attempted, mainly because the various relationships and dependencies are poorly understood.

Generation of synthetic prosody from text is limited by the capabilities of current text analysis algorithms. Natural prosody is a function of semantics (i.e. the meaning of the text), dialogue context, syntax, perception of rhythm, and human characteristics outside the realms of linguistic analysis (e.g. emotional state, age and state of health). For discourse-neutral prosody, syntactic and rhythmic analysis is probably sufficient, but long passages and spoken dialogue systems require semantic information for optimal performance. This discussion will concentrate on current methods for generation of discourse-neutral prosody, with the proviso that they should be capable of using semantic information appropriately.

The features considered here are prosodic phrasing, rhythm, intonation and segmental duration. Some progress has been made towards a unified framework for prosodic modelling, but there is still some way to go (this framework is based on metrical phonology, which is described in section 2.2). Prosodic phrasing and models of rhythmic and intonational phonology are represented in a single structure, although the initial phrasing can be developed independently. The durational model, although represented separately, depends heavily on information from this structure and has a strong metrical theme.

2.1 *Prosodic phrasing*

It is generally acknowledged that a sentence's prosodic tree is flatter than the syntactic tree. The classic example is the mismatch between the syntactic and prosodic structures for the embedded right-branching noun phrases (NP) in the sentence: 'This is, the dog that chased the cat that killed the rat,' which are respectively:

[S [This is [NP the dog that chased [NP the cat that killed [NP the rat...

[This is the dog] [that chased the cat] [that killed the rat...

Prosodic phrases divide the sentence into smaller units for easier processing, and provide an aid to syntactic disambiguation.

Early work by Chomsky and Halle [2] ascribed the flattening of the prosodic structure to 'performance', and deemed this area a linguistically uninteresting problem. This view is now generally discredited, and there is agreement that every utterance, in addition to its syntactic structure, has a prosodic structure of some type, and that there is a mapping between the two (although this mapping is not well defined).

The empirical evidence for prosodic structures is from two sources — psycho-linguistics and instrumental analysis. On the instrumental level, the existence of prosodic breaks, which are discrete events associated with acoustic cues (such as duration lengthening, pause insertion and intonation markers), provides evidence for prosodic phrase structure. Furthermore, these phrases would seem to be arranged hierarchically with boundaries dominating, many phrases being more prominent in some way. Prosodic phrase structure can be represented by different phrase break indices, the location and relative size of these defining the prosodic structure. However the acoustic cues are not always reliable, nor easily detected, and there is considerable debate as to how many 'levels' exist in a prosodic structure.

From a psycho-linguistic viewpoint, experiments indicate that intonation groups are planned in advance, since slips of the tongue tend to occur within these groups, and usually not at boundaries.

There are two main approaches to the prediction of prosodic structures - rule-based and stochastic.

Rule-based prediction

Many of the rule-based approaches stem from early work on performance structures based on experimental data, such as pausing and parsing values [3]. This work sought to account for the (then) disparity between linguistic phrase-structure theories and actual performance structures produced by humans, and focused on recreating the pause data of several analysed sentences from syntax (although they claimed that their method could easily account for other prosodic features). The central tenet of the work was that prosodic phrasing is a compromise between the need to respect both the linguistic structure and performance aspects of the sentence.

More recent efforts have extended the work on performance structure prediction to the prediction of prosodic phrasing. In this work, the basic rule-based approach is preserved, but other factors are introduced which are considered important for predictive purposes. For example, Bachenko and Fitzpatrick [4,5] believe that syntax plays a lesser role in determining phrasing, and that certain prosodic performance constraints, such as length, override syntactic structure. They allow prosodic boundaries to cross syntactic boundaries under certain conditions, whereas early work was essentially inter-clausal. Other modifications include counting phonological words rather than actual words when determining node strengths [5]. (A phonological word effectively functions as one spoken item, as the internal word-word boundaries are resistant to pausing [3]. Typical examples are determiner-noun word groups, such as the 'the + man'.) Further extensions incorporate punctuation into the predictive models [6], or assign more importance to specific features [7].

Stochastic methods

With the availability of large corpora, annotated with prosodic information such as location and salience of pauses, temporal information on durations, etc, the stochastic-based approach will come more to the fore. Recently methods for automatically predicting prosodic information using decision tree models have been described [8,9]. Generally, decision trees are derived by associating a probability with each potential boundary site in the text, and relating various features with each boundary site (e.g. utterance and phrase duration, length of utterance - in syllables/words - positions relative to the start or end of the nearest boundary location, etc) [9]. The resulting decision tree provides, in effect, an algorithm for predicting prosodic boundaries and their saliences (i.e. relative importance) for new input texts.

It is interesting to note that evaluations of both rule-based and stochastic methods show these methods attaining similar results [8].

Predicting prosodic phrase boundaries

There are essentially two main components in both of these models - determining potential boundary sites, and assigning boundary salience. This defines a prosodic hierarchy, with strong boundaries typically being associated with pauses. Boundary location is predicted in the rule-based method by grouping of words into phonological words, and then into phonological phrases, with reference to the syntactic structure for the sentence.

The stochastic approach determines potential boundaries by examining certain factors left to right, from the start of an utterance. An example is given in Fig 2 [9]. Note that this decision tree was tailored towards recognition of prosodic boundaries, to provide clues as to the syntactic structure for speech recognition.

if seconds from word on left to end > 0.0495455
then phrase boundary (99.7%)
else   if [no of seconds from last boundary]/[no of seconds from last phrase] < 0.601564
        then not phrase boundary (94%)
        else   if node is noun, noun phrase
                then not phrase boundary (88.6%)
                else   if word to left is not accented
                        then not phrase boundary (81%)
                else   if seconds from start to word on left < 2.49455
                        then not phrase boundary (75.3%)

Fig 2   Example decision tree for predicting prosodic boundary location.

Boundary salience is determined in the rule-based approach again with reference to the syntactic structure. A typical strategy is to balance material around the verbs in an utterance [4,5]. However, this is complicated by the tendency to equalise prosodic phrase size, at least in English. This can be accommodated either as the prosodic structure is derived, or by adjusting the final structures [10].

The stochastic approach determines boundary salience by reference to training data. For example, after a certain duration, or after a number of syllables or words, had passed since the last boundary was predicted, there would be a high likelihood of a salient boundary at the next potential boundary site.

A typical performance structure, predicted using a rule-based approach is given in Fig 3. The node values correspond to the number of phonological words they dominate, and translate directly into strong and weak boundaries, marked || and | respectively.

This performance structure may then be used as a basis for deriving metrical and rhythmic structures, intonation and duration.

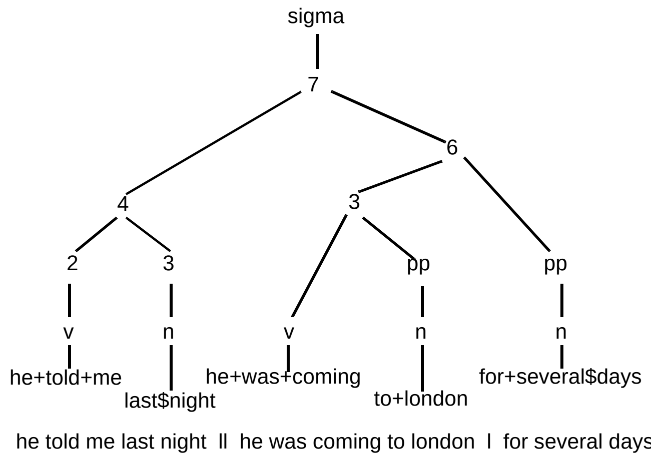

Fig 3   Prosodic structure tree.

2.2 *Rhythm and metrical structure*

One of the criticisms most often levelled at synthetic speech is that "...the rhythm doesn't sound right...". Since intonation and duration are the most significant correlates of perceived rhythm, their generation models should be linked to a common framework of rhythmic analysis. Metrical phonology [11] provides such a framework, and a suitable metrical structure can be derived from a combination of performance and syntactic structures.

The fundamental concept of metrical phonology [11] is that stress is represented in terms of the relative prominence of paired constituents. Only two stress levels are considered - s, meaning 'stronger than', and w, for 'weaker than'. These pairs may be at any prosodic level (syllable, foot, word, phrase, etc) giving a hierarchical binary tree structure with w s or s w labelling at each branching node. This is known as a metrical tree.

Stress and the metrical tree

The major advantage claimed for metrical trees is that they encode relative prominences of embedded constituents directly, without the need for cyclic application of stress rules. A problem arises, however, when dealing with more complex structures than a single pair. Consider the tree for the compound 'coffee table book', shown in Fig 4 (ignore the grid for now).

Using the w s relationship, 'book' is stronger than 'coffee table', and 'coffee' is stronger than 'table', but nothing can be said about the relative prominences of 'coffee' and 'book' since they are not sister constituents. The definition of a metrical head solves this problem. The metrical head is defined as being the stonger of a pair of sister constituents — hence 'coffee' is the metrical head of 'coffee table'. In this way, the *w s* relationship between 'coffee table' and 'book' is equivalent to a *w s* relationship between 'coffee' and 'book'. This allows relative prominences to be determined for any pair of strong constituents at the bottom level of the tree.

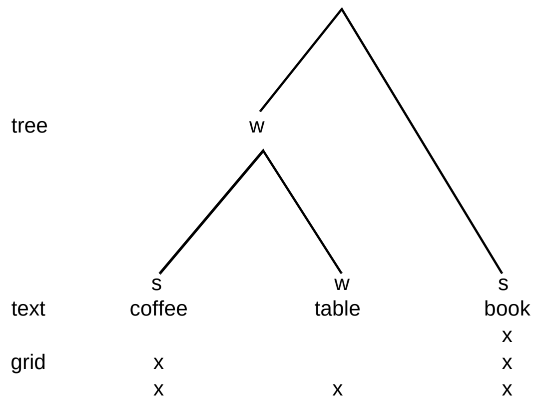

Fig 4 Example metrical tree and grid.

The metrical head any sub-tree, i.e. the constituent dominated only by strong nodes, is called the designated terminal element (DTE). The DTE of a sentence tree has special significance for assignment of phrase focus (nuclear accent) in terms of intonation.

Rhythm and the metrical grid

In many cases, relative prominences do not appear to be preserved under embedding in English. The classic example is 'thirteen men'. In isolation, 'thirteen' has a w s syllabic stress pattern, but appears to undergo a reversal to s w when embedded in 'thirteen men' (iambic reversal or stress shift). This can be viewed as a function of the rhythm of speech, and led to the definition of a linear structure for rhythm to complement the hierarchical representation of stress. This is the 'metrical grid', which shows relative prominences directly, as opposed to the tree which requires tracing of metrical heads to get equivalent information.

The grid is derived from the tree in three steps:

- assign a mark at level 1 on the grid to each terminal node,
- assign a mark on level 2 to each terminal node marked s,
- use the metrical head definition to assign marks at higher levels until all relative prominences are correctly represented in the grid (e.g. the DTE should have the highest prominence).

The last of the above steps is formally stated in the Relative Prominence Projection Rule (RPPR):

'In any constituent where the strong/weak relationship is defined, the DTE of its strong sub-constituent is metrically stronger than the DTE of its weak sub-constituent.'

This produces a minimal grid, representing the relative prominences with no redundant information. Referring back to Fig 4, the RPPR ensures that 'book' has a greater prominence than 'coffee'. Accents are assigned to 'coffee' and 'book', with the latter being focal, as it is the DTE of the tree. It is possible to extend the grid without affecting relative prominences by adding levels to certain constituents and re-applying the RPRR. This flexibility allows manipulation of the grid to satisfy rhythmic constraints, but has inevitably led to problems in interpreting the role of trees and grids in predicting stress and rhythm in speech. Three interpretations have been proposed. Tree-only phonology [11] contends that all essential stress and rhythm information is present in the tree. The converse holds for grid-only phonology [12], whereas tree-and-grid phonology [13] maintains that both structures are necessary with trees and grids representing stress and rhythm respectively. The dual approach has been developed mainly by Hayes [13], and seems to produce a more compact phonology with fewer ad hoc rules and constraints than either of the single-structure approaches.

The concept of eurhythmy is central to the metrical structure, a detailed discussion of which can be found in Hayes [13]. Basically, this assumes that human perception finds certain rhythmic patterns more pleasing than others, and this influences the rhythmic structure of spoken language. This phenomenon is most evident in highly structured language such as poetry. Various rules can be incorporated to improve eurhythmy, such as the Rhythmic Adjustment Rule (RAR), which generates alternating prominence patterns by transforming w w s sequences into s w s sequences under certain constraints. This is useful for noun phrases of the form 'adjective, adjective..., noun' which would have a default right-branching structure with the noun as the only strong element.

Building a metrical structure

Metrical analysis can be tied to a performance structure, as discussed in section 2.1. For example, given the locations and boundary strengths of phonological words and phrases, and possibly higher phrase levels (major, minor, intermediate, depending on the prosodic model), a binary tree can be constructed which represents the phrase structure. Where relative boundary strengths are not defined (within phonological words, and sometimes phrases), rules must be developed to expand the tree fully to word or even syllable level. The default for phrasal constituents is a right-branching structure with w s labelling at each branching node. In practice, the labelling for branching nodes is determined by a set of syntactic rules to give an initial tree and grid, which are then adjusted according to eurhythmic rules.

Having built a metrical structure, it can be used to guide the generation of an intonation pattern for the synthetic utterance.

2.3 Intonation

Ideally, any discussion of synthetic intonation for TTS should be preceded by a definition of the role of intonation in natural speech. Unfortunately, this is a complex task in itself. For example, the simple sentence:

John drove to London yesterday

could be the answer to at least four distinct questions depending on the intonation. Determining which intonation pattern is appropriate in a given dialogue context is a sophisticated text analysis problem. The intonation algorithms in most practical TTS systems are based on syntactic analysis at sentence level, giving discourse-neutral intonation. Some systems perform rudimentary semantic analysis for given/new information, contrastive stress and other linguistic factors spanning more than one sentence. However, the development of successful semantic analysis algorithms for TTS is still some way off. Consequently, current synthesisers for unrestricted TTS applications concentrate on optimising automatic discourse-neutral performance, with options to modify the intonation by direct annotation for dialogue applications.

With this in mind, two major requirements of an intonation component for TTS can be defined:

- pitch movements on accented words and at prosodic boundaries should closely mimic those of natural speech,
- accents should be distributed such that a natural rhythm is imparted.

The first of these has received the larger share of research effort to date, so that many systems have the potential to produce good intonation, but fail to do so, mainly due to simplistic accenting algorithms. This discussion will concentrate on the development of an intonational component which aims to satisfy both of the requirements above.

2.4 Basics of intonational modelling

The problem of generating synthetic intonation can be separated into two parts:

- the phonological component,
- the phonetic (or acoustic) component.

Intonational phonology provides a means of abstractly describing an intonation contour, whereas the phonetic component converts that abstract representation into a physically meaningful signal, e.g. a stream of frequency values (the pitch or F0 contour). Acoustically, the F0 contour appears continuous, and it is tempting to characterise it in terms of ‘shapes’ which may span whole phrases. However, the modern trend in intonational phonology is to specify a sequence of discrete tones, which are then joined by some interpolation process to form a continuous contour. These tones are usually of two types - pitch accents associated with prominent syllables, and boundary tones associated with varying degrees of prosodic boundaries (these may be derived from a performance structure as described previously). A widely used tonal sequence model was developed by Pierrehumbert [14]. A detailed understanding of tonal sequence models is not required in order to appreciate the essence of the intonational approach under discussion, but the salient points of the Pierrehumbert model are mentioned briefly for convenience.

A set of pitch accents is defined, based on two phonological levels — high (H) and low (L). The basic accents are labelled H* and L*, and can be combined to form bitonal accents with two-tone levels, e.g. H+L*, H*+L, etc (the * indicates the dominant tone, i.e. the one aligned with the accented syllable). Each accent is given a prominence on an abstract scale, which is eventually mapped to an F0 level within a specified F0 range. The model is essentially linear, in that each accent is specified independently, and the phonology does not cater for intonational dependencies at phrase level. One exception to this independence is intonational downstep, which is triggered by certain sequences of bitonals, where the prominence of an accent is a scaled version of the prominence of the preceding accent. As with phrase-level effects, there is no independent phonological control of downstep. These phonological constraints will be examined more closely later.

If a model can produce reasonable approximations to natural pitch contours, then the next step can be considered. This, is to determine suitable locations, types, prominences and F0 ranges for the accents and boundary tones by automatic analysis of the text. This task can be broken into three stages. Firstly, pitch accents are assigned at word level based on the metrical structure of the sentence. Secondly, a phonological model encompassing both accent downstep and phrase-level effects is imposed, which determines the type, prominence and F0 range for each accent. Boundary tones are not discussed in detail, as they are determined by the prosodic phrase structure. Finally, a phonetic model generates an F0 contour which can be imposed on the synthetic speech. The discussion will concentrate on the development of a phonological model which incorporates prosodic phrasing, rhythmic phonology and intonational phonology within a single metrical framework. The phonetic model used is that developed by Silverman [15] for Pierrehumbert's phonology, although other models are worthy of investigation (e.g. Taylor's rise-fall-connection model [16]).

Pitch accent assignment

For the discourse-neutral task, it is assumed that all the information necessary for natural accent placement can be derived by a combination of syntactic and rhythmic analyses of the text. Indeed, Hirschberg [17] asserts that good performance can be obtained on longer texts without the need for semantic, pragmatic and discourse information, by training classification trees using currently available text analysis techniques. Monaghan [18] uses a combination of accent over-generation followed by deletion rules to improve rhythm, whereas Dirksen [19] incorporates some aspects of metrical phonology to predict rhythmic structure on a more formal basis. All of these methods attempt to impart better rhythmic structure to synthetic speech, a goal completely ignored by the simple content/function word method used in many TTS systems. In that method, each content, or open class word (basically each noun, verb, adjective and adverb) is accented, and anything else is not.

Within the metrical framework, accents are assigned to words with grid levels of 2 or more (i.e. metrically strong words), as implied in the example of Fig 4. Although this is presented in the context of discourse-neutral intonation, the metrical structure can easily incorporate a priori accent placements (e.g. from stored patterns, text annotation, or full semantic analysis when available), and distribute the remaining accents to optimise rhythm.

Having built a tree as normal from the performance structure, it would first be labelled to ensure that each accented node is strong. In particular, if phrase focus is pre-assigned, the focal word should be made the DTE. Additional accents would then be derived by applying eurhythmic rules and constructing the grid, ensuring that the best rhythmic structure is obtained with varying degrees of a priori information.

A phonological model of intonation

Gradient variability — phonological models generally have a finite set of tonal levels. Unfortunately, examination of natural F0 contours suggests that F0 targets can take a continuous range of values unpredictable from the phonology alone. This has been termed gradient variability by researchers, and is considered by Ladd [20] to be: '...the most serious empirical weakness of a great many quantitatively explicit models of F0.'

In the phonetic realisation of F0 contours, Silverman's implementation [15] of the Pierrehumbert model uses two gradient parameters, prominence and range, which allow local and global F0 variability respectively. Although these make provision for gradient variability within phonologically equivalent contours, there is no reliable method of assigning meaningful values to them. In synthesis, a fixed alternating pattern of prominence patterns is generally applied, so that gradient variability in natural F0 contours is treated as essentially unpredictable.

On the other hand, Ladd [20, 21] has proposed another source of variability in F0 based on a single hierarchical structure to control accent downstep, register shift at phrase boundaries, and other effects previously ascribed to purely gradient variability. The structure is a binary tree for control of pitch register (a register tree), and can potentially produce a much richer variety of F0 contours. The attraction of this approach is that register trees bear a striking resemblance in both structure and philosophy to metrical trees, and can be derived in a principled manner from the metrical structure described in section 2.2.

Register trees — since downstep is a relational phenomenon, it only makes sense when applied to pairs of accents or phrases — hence the binary tree structure. Analogous to metrical trees, each binary pair can be only labelled as H-L or L-H. In this case, however, the labelling is not symmetric, as H-L denotes a register downstep, whereas L-H represents no change as opposed to upstep. The highest terminal element (HTE) of a tree or subtree is defined as the terminal node dominated exclusively by H nodes. To reinforce the analogy, Ladd linearised the tree such that a degree of downstep could be directly associated with each accent. This structure will be referred to as the register grid. The register grid is constructed by applying the Relative Height Projection Rule (RHPR):

- RHPR: in any constituent where the H L relation is defined, the HTE of the L subconstituent is:

  — one register step lower than the HTE of the H subconstituent when the L subconstituent is on the right,

  — at the same level as the HTE of the H subconstituent when the L subconstituent is on the left.

With an initial register defined, application of the RHPR defines the register movement throughout the sentence in terms of register steps. The shift between accents within tone groups (true downstep) may differ from that between tone groups (for example, complete register reset may occur between major intonational phrases). This gives considerable flexibility in modelling F0 contours for speech synthesis.

Implementation of a register tree model — a register tree can be derived by collapsing the metrical tree so that the domain of each terminal node contains one pitch accent. Labelling rules are then applied which favour downstepping sequences within tone groups, and across minor tone boundaries, but inhibit downstepping at major boundaries. The register level associated with each accent applies over the domain of the corresponding terminal node. Since the metrical tree is derived from a prosodically based performance structure, register domains correspond to prosodic domains at various levels, ensuring that register shift only occurs at prosodic boundaries. In this way, elements of performance structure, rhythmic phonology and intonational phonology are combined in one cohesive structure, built to satisfy the requirements of all three.

Having determined a register for each accent, an appropriate level (intonational prominence) must be assigned within that register. The simplest method is to assign fixed prominences to each accent type, e.g. the focal or nuclear accent could receive a higher prominence than the others. A more sophisticated approach would be to derive intonational prominences from the grid prominences, thus modelling another aspect of gradient variability.

Assigning accent types — a varied set of accents is useful when characterising F0 contours in natural speech, but predicting which accents are most appropriate for synthetic speech is a more difficult problem. Many models limit themselves to one accent (high) with interpolation, and some make use of a low accent, but obtaining consistent results with a larger accent set is not practical at present. In combining the Pierrehumbert accents with the register tree model, the following accent assignment has been found to work well:

- assign an H* to the first and last accents in a minor phrase,
- assign H+L* to intervening accents in downstepping minor phrases,
- assign a nuclear H* to the DTE in each major phrase.

This gives a natural downstepping contour within minor phrases, and the final H* signals the end of a tone unit with a definite upward pitch movement.

An example — Fig 5 shows the complete intonational structure generated for the sentence:

years ago a young bride took a dowry to her new husband

with accents from a simple content word method for comparison.

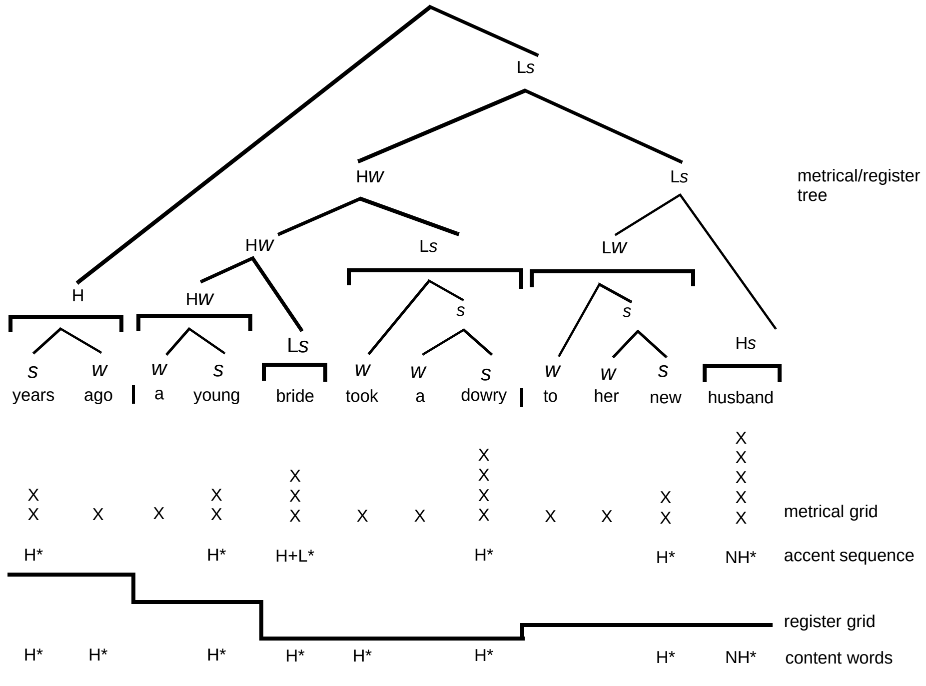

Fig 5 Example intonational structure.

Boundary tones are not explicitly shown in the accent sequence to avoid congestion in the diagram. Minor tone boundaries are indicated by the '|' character in the text string, and are of the fall-rise type L-H%, with a final fall L-L%. The tree is labelled for both metrical and intonational register relationships (w s and H L) at each node, illustrating the unified structure. Collapsing of the metrical structure gives the register tree, whose terminal nodes are denoted by horizontal bars. The bars also show the domain of the register associated with each terminal node. The metrical and register grids are built as previously described. This example illustrates several important features of the model:

- compared to the content word method, a more sparse accent distribution has been generated to satisfy rhythmic constraints,

- nested register shift between tone groups is evident — the last tone group is downstepped relative to the second, and both are downstepped relative to the first,

- because register shift is defined between HTEs, accent downstep within the second tone group results in an upward register shift at the boundary with the last group,

- although metrical and register trees share the same structure, they can have independent labelling strategies, e.g. w s does not imply H L.

The final stage of prosodic modelling is to predict segmental durations. By using information from the prosodic structure described above, the theme of a unified approach is preserved in the durational model.

2.5 Duration prediction

Natural speech clearly has a duration. It exists as an acoustic signal for a finite length of time. The rate at which it is produced may vary from the very slow to the very fast. When a speaker wishes to emphasise a given word or phrase, changing the duration of the speech signal is one of the techniques they employ. Natural speech has a particular rhythm to which we as humans appear to be very sensitive. Clearly synthetic speech, if it is to sound natural, must model the duration of a utterance.

The first challenge is to decide on an appropriate unit (quantum) of speech. That is the smallest descriptor of speech for which the term 'duration' can sensibly be applied. Clearly a word spoken in isolation has a duration and is a well-understood linguistic unit, but it cannot realistically be used as a quantum in duration analysis. Words have structure, they may be said to contain syllables and depending on the phonological analysis applied, phonemes or possibly tiers of features. In linear phonology, the phoneme and its associated acoustic realisation, the phone, act as convenient abstract quanta for durations. However, if the segmental approach imposed by linear phonology is rejected, then some other abstract unit of duration must be postulated which associates with bundles of features. The majority of work conducted on durations in speech have adopted the phone as the quantum for duration modification.

These units of speech are assumed in many statistical analyses of the speech signal to possess intrinsic or inherent durations, which vary due to a set of independent modification factors and hence have a characteristic probability distribution [22, 23]. The Gamma function, which has the property of describing random variables bounded at one end, has been shown to provide an acceptable model for the distribution of phone durations [24].

However, for computational expediency, many researchers use normal or log normal distributions to describe the behaviour of phone durations. The choice of distribution used by researchers depends upon the nature of the predictive model, i.e. whether the model assumes the factors affecting duration are additive or multiplicative. Such methods are exemplified in the classic model developed by Klatt [25] for the MITalk system. Techniques such as CART trees [26] and HMMs [27] represent alternatives to the additive-multiplicative models, and require fewer assumptions about the processes involved in duration changes. However, as a consequence of their simplicity such methods require large amounts of annotated data and inevitably suffer from the problem of sparse exemplar data. Comprehensive discussions on these topics may be found in van Santen and Olive [28] and Kornai [29].

All the techniques described above assume that the quantal unit of speech is being acted upon by a set of external factors which contribute to the observed duration. These factors include:

- phonetic identity,

- phonetic context,

- position within a syllable, word, phrase and sentence,

- speaking style,

- speaking rate,

- lexical stress,

- semantic prominence.

The power of probabilistic methods, such as HMMs, over predominantly deterministic methods, such as multiplicative models, is that much of the variance observed in the duration of segments can be accommodated within a sufficiently complex model, without the need for a clear understanding of how or what factors are affecting the quantal unit. CART tree-based methods represent a half-way house between probabilistic models, which have very few assumptions, and multiplicative models, which are very prescriptive. CART trees attempt to 'learn' the relative significance of a given set of factors.

Syllable-based duration models

A number of recent papers have suggested that the syllable rather than the phoneme is a more appropriate and theoretically robust structure for the prediction of durations. Such papers suggest that the durations of segments within a syllable can be simply predicted once the syllable duration has been derived [30,31]. These methods can be viewed as separating the phonetic identity and contextual effects from other prosodic effects.

The aim of phonological theory is not the prediction of segmental durations for synthesis; in fact many phonological theories would deny the existence of a phoneme, let alone have anything constructive to say about any associated acoustic duration. However, theories of phonology can be used to provide a skeletal structure on to which pragmatic prediction models can be fixed, and, if the theories are appropriately motivated, then the more sophisticated the skeletal structure, the simpler the prediction model needs to be. As a concrete example, if the structures of rhythm and prominence proposed by a number of phonological theories are correct, then, once constructed, they provide sufficient constraints to ensure that any production model will sound 'reasonable', provided it adheres to these constraints.

This assumption underpins a particular model under development at BT Laboratories (BTL) [32], which is based on predicting the duration of a syllable; it is suggested that the syllable represents a better unit for mapping to the acoustic signal than the complex and ill-defined mapping between the phoneme and the phone.

In this duration model the quantal unit of time is not associated directly with any phonetic element or context. It is an abstract quantity which resides on a 'skeletal' or 'timing tier' [33]. The quantal unit is constant and is not affected by the internal structure of the syllable. The prominence or salience of a syllable is, however, affected by the metrical structure of the utterance in which it resides. The stress-timed tendency of English clearly suggests that syllable duration is affected by metrical structure. Given that this method assumes a single quantal unit which is constant for all syllables, internal syllable structure must have an effect and the degree to which it affects syllable duration is dictated by the metrical structure of the utterance. One explanation of this is that there is a clash between the desire to conform to the constant beat on the one hand, and the need to satisfy phonetic and perception constraints on the other. Phonetic constraints include the phonetic identity of the phone while perception constraints consider the cues required to perceive the rhythmic structure in the language.

The phoneme durations predicted by this method still exhibit known problems; in particular, the method of deriving phonemes from the predicted syllable duration is currently over-simplified. As a consequence, phrase final syllables exhibit an asymmetry in the duration of phones in the onset and rime [34]. Any approach such as this succeeds or fails on the accuracy and completeness of the phonological information supplied by the other components of the text-to-speech system. For example, a lack of information concerning the degree of prominence of words in a phrase leads to inappropriately short durations being assigned.

Finally, one constraint seldom mentioned is the interaction between the duration prediction method and the speech synthesis technique. This effect is particularly significant for unit inventory-based synthesis techniques. The duration of a phoneme segment is only one of a number of factors leading to the perception of rhythm in synthetic speech. Factors such as the degree of vowel reduction, assimilation, elision and the intrinsic duration of unit elements can all adversely interact with a predicted duration.

At this stage in the text-to-speech conversion process, a description of the linguistic realisation of the original text has been constructed. The last remaining stage is to convert this symbolic representation into the final speech signal.

3. Speech generation

The process of generating synthetic speech from a linguistic structure involves a mapping from an abstract symbolic representation of language to a parametric continuous representation. This mapping represents a demarcation point between the processes involved in describing language and the processes involved in the production of speech. The parametric information is used to drive some form of speech production model.

Speech production methods can be broadly divided into two categories — those which predominantly model the speech signal and those which predominantly encode aspects of the speech signal. There are two prominent methods of modelling speech — 'articulatory synthesisers' and 'formant synthesisers'. Both generate a synthetic speech signal purely from parametric information driving an abstract model of production. Articulatory synthesisers attempt to produce synthetic speech by modelling the characteristics of the vocal tract and the speech articulators, while formant synthesisers attempt to produce synthetic speech by modelling characteristics of the acoustic signal. Alternatively, examples of the speech signal may be encoded by some process. Encoding schemes vary from the LPC-based methods such as impulse driven LPC, RELP and CELP to simple PCM coders [35].

By far the most successful method of modelling the speech signal is the formant synthesiser. This class of model has been shown repeatedly to produce good quality copies of natural speech, provided appropriate parameters for driving the model are obtained. In theory such models appear capable of reproducing most sounds used in speech. A number of commercially available systems are based on this technology -- for example, DECtalk. The problem with such models has always been in the specification of appropriate control parameters. Currently, most systems which use such models, while highly intelligible, still sound robotic. The 'holy grail' for such systems, is a generalised mapping from some symbolic representation of language to a set of parameters which produces highly natural-sounding synthetic speech. The ability of such models to copy speech accurately has encouraged some researchers to use these models as complex encoders. There are a number of advantages and some disadvantages with such an approach. The advantages are that much higher quality synthetic speech can be produced with comparatively low storage requirements, while retaining the ability to modify the encoded speech signal. The main disadvantage is that producing high-quality copy synthesis is an arduous and time-consuming process, making the production of a large database of speech sounds difficult.With the advent of cheap computer memory and more computationally efficient methods of manipulating and encoding the speech signal, such as PSOLA [36], the reasons for using a complex model are less clear. In reality the only significant advantage in using a complex formant model as an encoder is its potential for modifying the encoded signal.

The remainder of this discussion will concentrate on concatenative speech synthesis, in which units selected from a stored speech database are joined together, and artificial prosody is imposed upon the resulting speech sample. However, the unit selection techniques described could equally be used with other methods of speech generation.

3.1 Selecting speech units

The phoneme is still the most widely used symbolic representation of sound in text-to-speech systems. Depending on the phonological analysis applied, there are up to 44 phonemes in southern British English, and this set represents the minimum number of symbols required to uniquely describe any word spoken in that dialect. However, simply storing one example phone for each phoneme will not produce good quality speech synthesis. Processes, such as co-articulation, cause the production of one phone to be highly influenced by its neighbours. As an example consider the words 'spoon' and 'spin', which may be phonemically transcribed as /s p u n/1 and /s p I n/. There are a number of co-articulatory processes acting within these words. The vowels /u/ and /I/ are being affected by the preceding /p/ phone, making the onset to the vowels slightly aspirated, and by the following nasal /n/ making the vowels slightly nasalised. While the vowels, in turn, are affecting the preceding /s/ and /p/ phones and the following /n/ phone. In the word 'spoon', the lips are rounded in preparation for the production of the back rounded vowel /u/, while in the word 'spin' the lips are retracted in preparation for the production of the front unrounded vowel /I/. Such co-articulation effects highlight the need to store phones in context.

The unit inventory methods of speech synthesis generate synthetic speech from a finite set of example sounds. Typically a unit is based around some combination of phones. The fundamental limitation of such systems is the number of example sounds that can be stored. The first and still one of the most common units of speech used in speech synthesis is the diphone. The diphone represents a computationally expedient compromise between the most immediate effects of co-articulation and the desire to reduce the number of units stored. Diphones make two fundamental assumptions about the speech signal, both of which are known to be wrong but are used as pragmatic approximations (see Fig 6). The first assumption is that phones may be considered to be composed of an onset component, a steady-state middle component and an offset component. The second assumption is that the effects of co-articulation can be captured within the transition from one phone to another. The limitations of these assumptions can be seen by reference to the brief description of co-articulation given above. A diphone is defined as containing the transition from the steady-state portion of one phone to the steady-state portion of its immediate neighbour. A better approximation to the underlying speech signal can be obtained by defining a unit that contains the transitions from steady-state portions of neighbouring phones into and out of a phone. Such units are called triphones.

For the southern British English dialect described above which has 44 phonemes and silence, 1848 diphones and 79 422 triphones are required to cover every combination of phonemes. The triphone figure is a large overestimate, as many phoneme combinations are simply impossible in this dialect. A more representative estimate can be obtained by restricting the centre phoneme to one of the 22 vowels present in the dialect. This approximation results in a total of 38 892 triphones — still more than an order larger than the simple diphone set.

[1] Transcribed using the SAM-PA machine-readable symbol set [1].

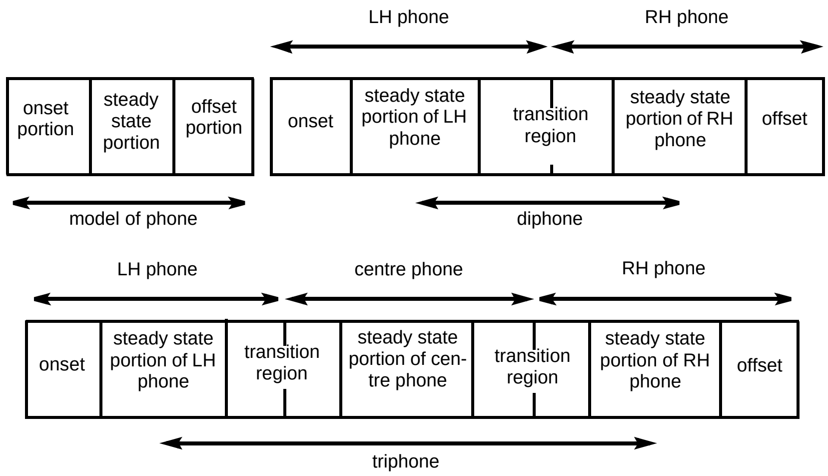

Fig 6 Underlying model of N-phone units.

Neither the diphone or the triphone are linguistically motivated units, as no account of the underlying structure of language is considered. This has led researchers to examine the syllable as an alternative unit. The proposition is that retaining the syllable structure improves the naturalness of the synthetic speech. However, a database which contained all legal syllables of a dialect and the syllable transition glue would be substantially larger than either a diphone or triphone inventory. A compromise solution proposed by a few researchers has been to develop a unit called the demi-syllable. Such units exhibit some of the properties of the syllable but are much reduced in number [37].

The search to find a set of units which represents an acceptable compromise between the desire to store as few units as possible while maintaining high quality synthetic speech, has led to the introduction of the non-uniform unit or N-phone unit [38] systems. The method of non-uniform unit selection involves searching a database of annotated speech for the most appropriate unit sequence. The eventual sequence may contain phones, diphones, triphones or larger units, depending on the minimum cost solution available in the database. N-phone unit systems contain a pre-selected set of units where the size of the unit reflects the acoustic complexity of the cluster of phones to be stored.

The previous discussion has assumed that the phoneme is the most appropriate label for the process of unit selection; there are, however, alternatives. Recently, researchers have been investigating other representations of speech data to supplement the simple phonemic description, but the idea that the phoneme label is insufficient is not new. Systems based on formant models have been applying phonetic rewrite rules for many years [39], and more recently nonlinear phonology has been applied to the specification of parameters for formant synthesis [40]. It is only with the growing awareness of the significance of prosody that unit-based systems have started to re-examine the sufficiency of the phoneme. Campbell and Black [41] have proposed supplementing the phonemic label with prosodic information, e.g. the type of intonation contour associated with a phone.

A phone carries with it a set of intrinsic characteristics. These characteristics are determined at the time of the recording, where the unit is subject to all the linguistic and paralinguistic forces normally present in natural speech. The significance of the recording has been known for some time. Researchers are constantly striving to define a controlled recording environment which produces units with the best compromise of intrinsic characteristics. One commonly used technique is to embed the unit within a carrier phrase, for example, ‘say bat again’, ‘say bit again’, etc. Using this technique database developers can more readily control the degree of emphasis on a word. Poorly controlled word emphasis can lead to overly articulated speech synthesis.

Increasing the amount of information on any one phone obviously adds substantially to the complexity of the unit selection process and blurs still further the minimum requirements of a unit inventory system. The danger is that with an increase in information about the environment of a given phone there will be an associated increase in the number of phones to be stored.

In the end, however complex or sophisticated the unit selection process, such systems represent a compromise between the specification of linguistic information and the size of the unit inventory. A question still remains as to whether some form of compromise can be found that will satisfy all the requirements of a unit inventory. If not, then signal manipulation will prove necessary to supplement missing environments.

3.2 Concatenation of speech units

In a concatenative text-to-speech system, once a set of speech units have been selected from a stored speech database, they must be joined to form a complete utterance. Since the units are selected from throughout the database, techniques are required to ensure that the resulting synthetic speech is free from sample-to-sample discontinuities and discontinuities in the pitch period, which would result in unnatural sounding speech, and would cause difficulties in the prosody modification stage (see section 3.3).

The simplest method of joining two units is to merge them together, by 'ramping' from one unit to the next, on a sample-by-sample basis. This simple joining process works reasonably well when one or both of the joining sections of speech are unvoiced. However, for joins in which both sides are voiced, it is necessary to employ a more sophisticated technique. One method of improving on the basic merging technique is to find the overlap giving the maximum correlation between the two signals, over the range of a single pitch period, and then carrying out the merge.

There are a number of problems with this method. Firstly, it is necessary to know the pitch period of one or both of the joining speech sections in order to decide on the range for carrying out the cross-correlation. Although it is reasonably easy to visually estimate this from a speech waveform, software methods which attempt to do this are prone to being unreliable and computationally complex. Secondly, the effectiveness of the method is dependent on how well matched the two signals are in terms of pitch period. If the pitch of the two speech sections is significantly different, the match achieved by the correlation process will not be very good. The overlap used for the merge must be limited to a single period, to minimise this pitch mis-match effect. This can result in relatively sudden changes in phoneme timbre in the speech after the joining process.

A solution to some of these problems is to synthesise segments from each side of the join in a manner which is pitch synchronous with the opposite side of the join, and then to carry out the merging process with an overlap which spans several pitch periods, shown in Fig 7.

This pitch synchronous approach solves the problem of limited overlap and also that of pitch period mismatch. However, it still requires that the pitch periodicity of each of the voiced sections is identified. In order to avoid having to carry out this computationally intensive task during the synthesis process, some systems make use of techniques which analyse the pitch characteristics of the entire speech database. This is carried out as an off-line process, which is only required once for each recorded speech database. For all recognisable voiced sections within the speech database a marker is created for a particular point in the pitch period. Using this information, it is possible to carry out the previously described pitch synchronous joining process in a computationally efficient manner.

3.3 Pitch time and 'quality' modification

The concatenation technique described above will not generally result in speech with natural sounding prosody. This is because the units that are used to construct the synthetic speech are selected from throughout the stored speech database, on the basis of phoneme content and context. Prosodic characteristics are not a consideration in the unit selection process, so the resulting concatenated speech will have an apparently arbitrary prosodic pattern.

In order to produce a more acceptable sounding speech, it is necessary to apply synthetic prosody in the form of pitch, loudness and segmental duration. Each of these can potentially be controlled throughout the speech synthesis process, but in practice loudness is found to be the least significant in terms of perceived prosody [42]. The rest of this section will concentrate on techniques for modifying the pitch and phoneme duration of synthesised speech.

Time-domain pitch-synchronous processing techniques

Techniques of this type make further use of the pitch-synchronous overlap-add method used for joining units, but act on the complete speech signal. The definitive paper on such techniques was written some time ago [36]. The inputs to this process are the speech signal (and its pitchmark information), the boundaries of the phonemes within the speech along with their required duration, and the required pitch contour. The pitch contour consists of a series of frequency values, corresponding to the required pitch at constant intervals throughout the speech. The duration and pitch modification tasks may be carried out in two separate steps, or as a single operation.

Duration modification — the speech waveform is decomposed into a sequence of short-term overlapping signals, which are obtained by multiplying the signal by a series of raised cosine windows centred on the pitchmarks. A new set of pitchmarks are calculated. The number of pitch periods in the new speech signal is modified to give the required duration. The spacing of the pitchmarks is derived from the original sample. The short-term signals are overlap added, centred on the new set of pitchmarks, to produce a new speech signal. Unvoiced sections of speech, which do not contain pitchmarks, first have artificial pitchmarks added, at constant spacing, before being processed in the same way. In the resulting speech, pitch periods from within the original speech signal are either duplicated (when increasing the phoneme duration) or omitted (when

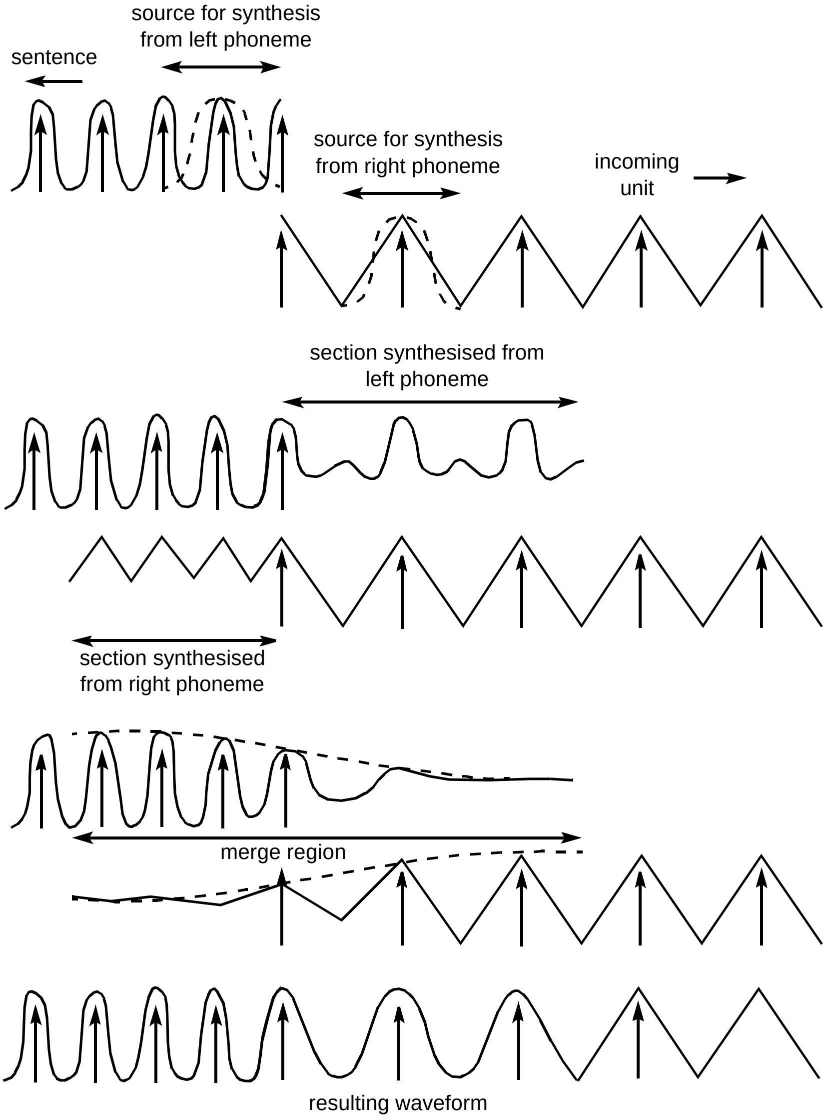

Fig 7 Pitch synchronous technique for joining voiced units.

decreasing the phoneme duration), according to a mapping function.

Pitch modification — the speech waveform is decomposed into a sequence of short term signals in a similar way to the duration modification stage. However, the new set of pitchmarks are generated according to the required pitch, as defined in the pitch contour input to the prosody modification process. The resultant speech signal is then formed using the overlap-add process previously described. An example of both pitch and duration modification of a speech signal is shown in Fig 8.

Linear prediction techniques

Although time-domain overlap-add techniques are computationally efficient, they are known to introduce audible spectral distortion. Weighted averaging of time-shifted speech signals (or correlated signals in general) results in phase distortion, and has been shown to broaden formant bandwidths, particularly at higher frequencies [36]. If, however, the signal to be modified is uncorrelated and spectrally flat, phase distortion is less relevant as the phase is unstructured to start with, and there are no spectral peaks in the envelope. In speech signal processing, linear prediction (LP) is a widely used technique for separating the signal into a spectral envelope component and a spectrally flat residual or excitation component. By applying pitch and duration modification algorithms to the residual and then recombining with the envelope, distortion around spectral peaks can be reduced.

In the source/filter model of speech production, the LP residual and envelope signals can be roughly interpreted as the source and vocal tract components respectively. In practice, these components are not completely independent, and pitch modification should ideally be coupled to vocal

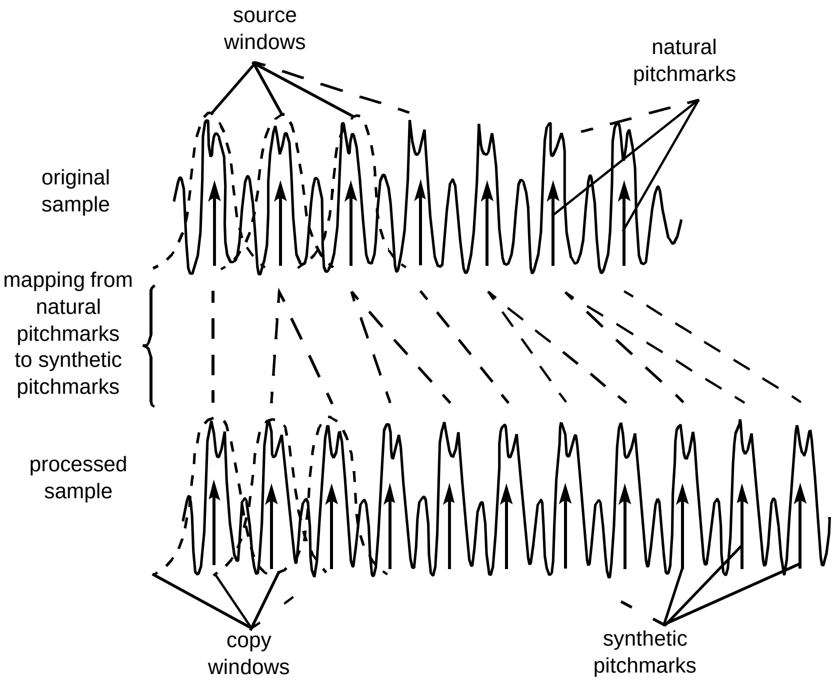

Fig 8 Example of prosody modification using a pitch synchronous technique (duration and pitch increased).

tract modification to model the process more accurately. This interaction tends to play a more dominant role in higher pitched female and child voices, which suffer more degradation from time-domain methods than low-pitched male voices. Techniques based on vocal tract/source separation may therefore prove more robust over a wide range of voices, provided the vocal tract modelling is sufficiently good.

A particular method developed at BT is based upon pitch-synchronous resampling of the residual [43]. Smooth resynthesis is ensured by interpolation of filter parameters at each sampling instant between frame centres during analysis and synthesis to avoid transients. The pitch period is modified by resampling a portion of the residual corresponding roughly to the glottal open phase, so that pitch modification is achieved without recourse to overlap-add. In this way, high frequencies injected at glottal closure are retained, as is the overall shape of the residual signal, and modifications are limited to the perceptually less important open phase of the glottal cycle. To further maintain smooth signal characteristics, the resampling factor is not constant over the modified section, but varies according to a sinusoidal function.

4. Conclusions

The conversion of text to speech has been presented as a two-stage process of analysis and generation. This paper has described some techniques used in the prosody and speech generation stage of current state-of-the-art commercial TTS systems.

It is widely held view that the greatest weakness of current TTS systems is in generating natural sounding prosody, and it is the authors' contention that improving the overall rhythmic quality of synthetic speech is the key to solving this problem. Metrical analysis was described as providing a common structural framework for the generation of synthetic prosody, in particular for accenting, so that rhythmic characteristics are maintained.

Current high-quality TTS systems all employ speech concatenation-based methods for generating the speech signal. This paper has described a framework for the selection of appropriate speech units from a pre-recorded database, and an algorithm for smoothly joining them. It is then necessary to impose synthetic prosody on to the speech, and a popular time-domain method was described.

However, there are a number of limitations to such methods, particularly the introduction of phase distortion and the inability to control the spectral shape of the resultant signal. A number of methods have been developed to avoid some of these pitfalls by separately modelling the spectral envelope of the speech, allowing finer control of the spectral shape in the final synthetic speech.

The perceived quality of synthetic speech has shown a significant improvement in the last decade, to the point where it is considered acceptable for many applications, such as telephone-based information provision. While there are many problems still to solve in TTS research in analysis, prosody and speech generation, the next ten years may well see a shift away from straightforward text-to-speech conversion towards more ‘intelligent’ speech synthesis systems. These systems will be designed to interact with a dialogue model, and will use text as an intermeidate representation between an underlying machine ‘intention’, and the synthetic spoken output. This approach will require a deeper understanding of issues such as controlling of speech style, and the communicative function of prosody. It will then be possible to build machines that ‘communicate with’ the user, rather than just ‘talk to’ them.

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43 European patent No 94301953.9.

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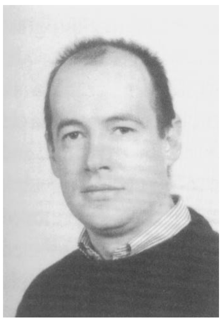
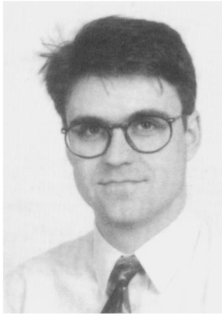
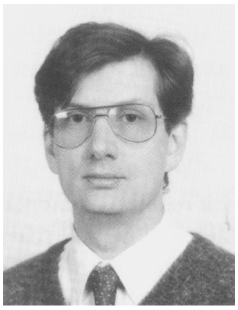
He joined the speech synthesis and analysis group at BT Laboratories in 1993, where he has been involved in many aspects of text-to-speech synthesis research and development, in particular, morphology, pronunciation and multilingual issues. His current research interests include speech modification algorithms and the use of semantic information in TTS. He is a Member of the IEEE and an Associate Member of the IEE.



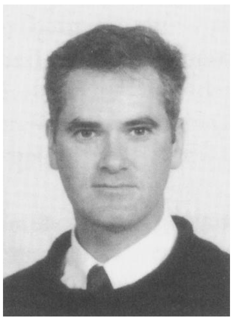
Andrew Lowry received the BSc and PhD degrees in electrical and electronic engineering from Queen's University, Belfast in 1984 and 1988 respectively. He joined BT Laboratories in 1988 to work on format analysis and synthesis of speech, and has been primarily involved in developing signal processing algorithms for various aspects of text-to-speech synthesis. Current research interests include intonation modelling for TTS, and speech analysis using nonlinear dynamics and higher order statistics. He is a Chartered Engineer and a Member of the IEE.



Peter Jackson graduated from UMIST with an MSc in electrical engineering and electronics in 19983. He then joined BT Laboratories, where he initially worked on the electronic design of consumer products. He has also worked on development of the BT range of modems and in the area of speech applications. His current work includes the development of prosody modification and joining algorithms for TTS.



Andrew Breen holds a BSc in Physics with Computing Physics from University College Swansea, an MSc (Eng) in Electrical Engineering from Liverpool University and a PhD in Speech Science form University College London. After completing his first degree he was employed in the department of Electronics and Electrical Engineering at Liverpool University where he worked for three years in the area of speech coding and digital filter design. After leaving Liverpool he was employed as an experimental officer in the department of Phonetics and Linguistics at UCL where he worked on speech synthesis and analysis. On joining BT in 1989 he worked for two years on continuous speech recognition before moving into the area of text-to-speech synthesis. He is project leader of the language component of the Laureate TTS system and project manager of the Synthetic Persona project. His current research interests include speech synthesis, speech prosody and the representation of linguistic and para-linguistic information in a synthetic persona. He is a member of the IEE.



Steve Minnis joined BT's software engineering division in 1987 after graduating with an honours degree in computer science at the Queen's University of Belfast. He was initially involved in software design methodologies and software quality metrics. Later, a spell as a contractor working on machine translation in Japan, at Mitsubishi electric research laboratories, led to an interest in natural language processing. He rejoined BT in 1992, in the newly formed language group, looking at parsing for text-to-speech, and parsing techniques in general.

His current work is research new methods for generation of appropriate prosody for text to speech. Other research interests include lip-motion synthesis for automatic talking heads, and sign language translation. He is currently studying for a part-time PhD in prosodic parsing at York University.